# Stage 3: Supervised Learning and System Integration

Final Report — Pipeline, Classifiers, and In-Class Challenge Results

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**Repository:** github.com/ekw4tu/DS3001-Project
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# 1. System Overview

Our Stage 2 + Stage 3 pipeline is packaged as a single Python project under `src/` (importable library) and `scripts/` (CLI entry points). The original Colab notebooks are preserved verbatim in `notebooks/`; a TA can either open the notebook in Colab or run `scripts/evaluate.py` to reproduce every metric on the held-out test set with one command.

### End-to-end flow

```
data/Gallery, data/Probe
       │
       ▼
walk_image_tasks ──► InsightFace detect + align ──► ArcFace (512-d) ─┐
       │                        │                                    │
       │                        └─► bbox crop + VGG19 fc2 (4096-d) ──┤
       ▼                                                            ▼
parse_base_identity / parse_condition                       stage2_embeddings.pkl
       │                                                            │
       ▼                                                            ▼
stratified 80/20 split (identity × condition) ──► train_embeddings / test_embedding
       │
       ├─► Condition LR (ArcFace, VGG19)
       ├─► Identity LR (ArcFace, VGG19)
       ├─► VGG19 L2 + PCA + LR
       ├─► VGG19 MLP head (frozen backbone)
       ├─► VGG19 SupCon contrastive head ──► nearest-centroid (unit sphere)
       └─► ArcFace gallery + identify() cosine ──► gallery_embeddings.pkl
```

### Design choices

- **Shared face crop for apples-to-apples comparison.** VGG19 has no face detector; we reuse the InsightFace bounding box so both feature sets see the same pixels. Without this the two extractors would be on different crops and comparisons would be meaningless.
- **Data-driven identity parsing.** `parse_base_identity(folder, filename)` reads canonical labels from a token registry (`src/identities.json`) with a numeric-stem fallback so classmate folders `1Gallery`… `38Gallery` parse automatically without 38 registry entries.
- **Stratified split on identity × condition.** Ensures every (person, condition) cell lands in both train and test, so no bucket is silently untested.
- **Reproducibility via env vars.** `FR_DATA_ROOT` / `FR_ARTIFACTS_DIR` let the same code run on a laptop, a Mac, or Colab with Drive mounted.

# 2. Identity and Condition Classification

Rubric §1 · 14% of overall grade · all numbers on the 20% held-out test split (random_state = 42)

## 2.1 Condition Classification (Logistic Regression, 4%)

A class-balanced logistic regression over each feature set predicts one of five conditions (`clean`, `clean_probe`, `Expression`, `Lighting`, `Occlusion`, `Side`). `clean` and `clean_probe` are merged into a single *Clean* bucket for scoring so the model isn't penalized for splitting on the Gallery/Probe axis.

| Feature set | Overall | Clean | Expression | Lighting | Occlusion | Side |
|---|---|---|---|---|---|---|
| ArcFace | 0.545 | 0.656 | 0.462 | 0.375 | 0.833 | 0.500 |
| **VGG19** | **0.612** | 0.538 | 0.500 | 0.559 | **0.917** | **0.857** |

**VGG19 wins on condition.** This is the opposite of identity (where ArcFace wins — see §2.2). The reason is structural: ArcFace's detection + alignment pipeline *deliberately strips out* rotation, pose, and lighting so that faces map to canonical 112×112 tiles — those are the exact signals a condition classifier needs. VGG19 receives the raw bounding-box crop and retains orientation + lighting information in its fc2 activations, so an LR on top can recover the condition.

**Failure analysis (condition)**

- **Lighting vs Clean** is the dominant confusion for both models. Many "Lighting" probes are only mildly under-exposed and visually look clean; the LR collapses the boundary. VGG19's 0.559 on Lighting is already a lift over ArcFace's 0.375, consistent with VGG retaining luminance signal.
- **Occlusion and Side are easy** (>0.83 and >0.50 on ArcFace, >0.91 on VGG) because sunglasses and profile shots change the global image statistics dramatically.
- **Expression is the hardest middle class** — expressions don't shift gross pose or lighting, so the model has only subtle facial-action signal to work with. This is a feature-set limitation, not a tuning issue.

**Train vs test gap:** the class-balanced LR trains on ~480 samples across 5 classes. Training accuracy saturates near 0.95 while held-out is 0.55–0.61 — a ~35 pt gap. That gap reflects the small-data / high-dimensional regime (512-d or 4096-d features vs ~100 samples per class), not an architectural failure. Adding `class_weight='balanced'` prevents the LR from defaulting to the majority "clean" class.

## 2.2 Re-Identification Fine-Tuning on VGG (10%)

The rubric asks us to improve VGG re-ID over the Stage 2 baseline and compare against the easier "LR on top" and the harder "finetune VGG layers" paths. We implemented both, plus a third **SupCon contrastive head** that directly targets the rubric phrase "*clustering / distance-based identification*".

| Classifier | Type | Test accuracy | Notes |
|---|---|---:|---|
| ArcFace + LR (baseline comparison) | Supervised | **1.000** | Perfect on 10-identity closed set |
| ArcFace cosine `identify()` | Distance-based | **1.000** (97/97) | Production classifier |
| VGG19 LR raw (Stage 2 baseline) | Supervised | 0.810 | Rubric baseline |
| VGG19 L2 + PCA(256) + LR | Supervised | 0.785 | "LR on top" path from rubric |
| VGG19 MLP head (frozen backbone) | Supervised fine-tune | 0.818 | 3-layer head + BN + dropout |
| **VGG19 SupCon + nearest-centroid** | Contrastive fine-tune | See `evaluate.py` step [7] | Our answer to the "clustering / distance" prompt |

### Why three fine-tuning variants

- **L2 + PCA + LR (0.785).** The simplest "LR on top". L2 makes Euclidean distance approximate cosine; PCA-256 keeps 98.4% variance and kills per-feature noise. It does *not* beat the raw-feature LR because the 10-class regime is already well-separated and dimensionality reduction throws away discriminative dimensions.
- **MLP head (0.818).** VGG19 frozen, 3-layer head (4096 → 512 → 256 → 10) with BN + dropout (0.4, 0.3) trained with cross-entropy. Keeps the 140M VGG19 parameters fixed because our dataset is ~600 images — unfreezing VGG19 immediately overfits. The head delivers a modest lift (+0.008) over raw LR; cross-entropy doesn't reshape the embedding geometry itself.
- **SupCon contrastive head.** Architecture: `VGG19(frozen) → L2 → Linear(4096, 512) → BN → ReLU → Linear(512, 128) → L2`. Trained with Supervised Contrastive Loss (Khosla et al. 2020, temperature 0.1). Inference is nearest-centroid cosine on the 128-d unit sphere — exactly the geometry the rubric's "clustering / distance-based identification" language asks for. Cross-entropy only separates classes at the decision boundary; SupCon explicitly *pulls same-identity*

*embeddings together* and *pushes different-identity embeddings apart*, which is the property a cosine-NN or k-means lookup exploits.

## 2.3 Failure Case Analysis

### Per-condition identity accuracy (held-out test set)

| Condition | ArcFace | VGG19 raw LR | Gap (Arc – VGG) |
|---|---|---|---|
| Expression | 1.000 | 0.955 | +0.045 |
| Lighting | 1.000 | 0.882 | +0.118 |
| Occlusion | 1.000 | 0.750 | +0.250 |
| Side (profile) | 1.000 | 0.714 | +0.286 |
| Clean (gallery) | 1.000 | 0.762 | +0.238 |
| Clean (probe) | 1.000 | 0.667 | +0.333 |

**Where VGG19 fails (and why ArcFace doesn't).** The VGG gap is largest on *Side* (0.286) and *Occlusion* (0.250) — exactly the conditions that shift face geometry most. Without alignment, VGG19's fc2 activations are dominated by the pose/occlusion signal, so a profile shot of Nico lands closer to *any* profile shot than to a frontal Nico. ArcFace's pre-embedding alignment warps every face to the same canonical 112×112 template, so pose and occlusion are factored out before the 512-d vector is computed. This is the architectural reason ArcFace reaches 1.000 and the strongest case against the "VGG for identity" approach.

**Rubric note on the 75–80% target.** The rubric allows for an explanation if performance is hard to achieve, *provided we demonstrate progress from the vanilla baseline*. Our VGG MLP head (0.818) and L2+PCA+LR (0.785) both *meet or exceed* the 75–80% range, and the SupCon contrastive head is the principled "distance-based" variant the rubric explicitly asks for. ArcFace at 1.000 confirms the ceiling is not the data but the feature set.

### Train vs test

VGG LR trains to ~0.99 and tests at 0.81 — an 18 pt gap. The gap *decreases* with the MLP head (0.95 / 0.82) because BN + dropout regularize. ArcFace LR trains and tests at 1.0 — the features are already identity-separable, so the LR has nothing left to overfit.

### Where to look in the code

| Artifact | Location |
|---|---|
| Condition LR | `src/train.py::train_condition_classifier` |

| | |
|---|---|
| Identity LR | `src/train.py::train_identity_classifier` |
| VGG L2+PCA+LR | `src/train.py::train_vgg_pca_pipeline` |
| VGG MLP head | `src/train.py::train_vgg_mlp_head` |
| VGG SupCon head | `src/train.py::train_vgg_contrastive_head` |
| ArcFace gallery + identify() | `src/identify.py` |
| Reproducer | `scripts/evaluate.py` (one command runs all 9 steps) |

# 3. System Documentation and Reproducibility

Rubric §2 · submit before 4/29 for 5% multiplier

## 3.1 Installation

```
git clone https://github.com/ekw4tu/DS3001-Project.git
cd DS3001-Project
pip install -r requirements.txt
```

First run of `insightface` downloads the `buffalo_l` ONNX weights (~280 MB) to `~/.insightface/models/`. Subsequent runs hit the cache.

## 3.2 Data layout

```
data/
├── Gallery/
│   ├── willGallery/        willSmithGallery1.jpg, ...
│   ├── kimGallery/         ...
│   └── <personGallery>/    ...
└── Probe/
    ├── willProbe/
    │   ├── willExpression/   ...
    │   ├── willOcclusion/    ...
    │   ├── willLighting/     ...
    │   └── willSide/         ...
    └── <personProbe>/<condition>/*
```

Any folder whose name contains a known identity token (see `src/metadata.py`) is picked up automatically. Classmate folders named `1Gallery` … `38Gallery` parse by numeric stem fallback and do not need registry entries.

## 3.3 One-command reproduction for a TA

```
# Extract all features + build gallery + write train/test pickles
python scripts/build_gallery.py --gpu

# Reproduce every Stage 3 metric on the held-out test set
python scripts/evaluate.py
```

`evaluate.py` prints, in order:

1. ArcFace → Condition (class-balanced LR)
2. VGG19 → Condition (class-balanced LR)
3. ArcFace → Identity (LR)
4. VGG19 → Identity (raw LR)
5. VGG19 → Identity (L2 + PCA + LR)
6. VGG19 → Identity (MLP head over frozen backbone)
7. **VGG19 → Identity (SupCon contrastive head + nearest-centroid)**
8. ArcFace `identify()` cosine validation
9. Per-condition identity accuracy breakdown

## 3.4 Adding a new identity (no full rebuild)

```
# 1. Drop gallery photos into data/Gallery/<personGallery>/
cp /wherever/alice_*.jpg data/Gallery/aliceGallery/

# 2. Append to the gallery in seconds
python scripts/add_identity.py \
    --name "Alice Smith" \
    --folder data/Gallery/aliceGallery \
    --match-tokens alice smith

# 3. (Optional) rebuild full embeddings only if you want alice in train/test splits
python scripts/build_gallery.py
```

## 3.5 Colab path

Open `notebooks/Colab_Quickstart.ipynb` or `notebooks/Challenge_Day.ipynb` in Colab, run top to bottom. Mounts Drive, sets `FR_DATA_ROOT`, calls the same scripts as above. GPU for free on a T4 or A100 runtime.

## 3.6 Environment variables

| Variable | Effect |
|---|---|
| `FR_PROJECT_ROOT` | Override repo root (normally auto-detected) |
| `FR_DATA_ROOT` | Override data path (Colab: `/content/drive/MyDrive/DS_NN_Project_Pictures`) |
| `FR_ARTIFACTS_DIR` | Where to save `.pkl` / `.pth` outputs |

# 4. In-Class Challenge: Unknown Identity Matching

Rubric §3 · 6% · 4/21 session · 5 unknown feature vectors, 3 correct = full credit

## 4.1 Input format

The TA distributed 5 `.npy` files, each a dict-wrapped single vector:

- `ID1.npy`, `ID2.npy`, `ID3_UPDATED.npy` — **512-d** ArcFace (InsightFace `buffalo_l`)
- `ID4.npy`, `ID5_UPDATED.npy` — **1000-d** VGG19 **pre-softmax logits** (not the 4096-d fc2 we use by default)

Classmates were photographed in-session (1–3 photos each) and added to Drive as `1Gallery/`… `38Gallery/` folders alongside the 10 original celebrity folders. Total gallery after ingestion: **~180 vectors across 48 identities**.

## 4.2 Route-by-dimensionality workflow

```
# Sniff dim for each uploaded .npy
import numpy as np
arr = np.load(path, allow_pickle=True)
if arr.dtype == object: arr = arr.item()['features']  # TA's dict wrapper
print(arr.shape)   # (512,) → ArcFace, (1000,) → VGG logits
```

**ArcFace unknowns (512-d)** — cosine match against the full gallery via `scripts/identify_unknowns.py`:

```
python scripts/identify_unknowns.py /content/ID1.npy --top-k 5
python scripts/identify_unknowns.py /content/ID2.npy --top-k 5
python scripts/identify_unknowns.py /content/ID3_UPDATED.npy --top-k 5
```

**VGG unknowns (1000-d pre-softmax logits)** — new `scripts/match_vgg_logits.py` extracts matching 1000-d logits from the gallery on-the-fly (VGG19 `fc2 @ W + b` using the `predictions` layer's own weights) and cosine-matches:

```
python scripts/match_vgg_logits.py /content/ID4.npy /content/ID5_UPDATED.npy \
    --classmates-only --center --max-sim --gpu --top-k 5
```

`--classmates-only` drops the 10 celebrity identities so dense celebrity centroids don't drown out sparse classmate centroids. `--center` subtracts the gallery-mean logit (removing the common "face-ish" ImageNet component that squeezes every cosine into 0.70–0.85). `--max-sim` scores each identity by their *closest* gallery image, not the centroid — preserves the one good shot when classmates have only 1–3 photos.

## 4.3 Source-level patches merged for the challenge

Three bugs surfaced during class and were fixed permanently:

- `walk_image_tasks` previously `break`-ed after the first of `Gallery/` / `Gallery /` was found, silently dropping the other. Now walks both with dedup via a `seen` set.
- `parse_base_identity` had no entry for numeric folder stems, so `1Gallery` … `38Gallery` all parsed to `"Unknown"`. Added a numeric-stem fallback.
- `identify_unknowns.py` couldn't read the TA's object-dtype dict-wrapped `.npy` files (`{'features': array(...)}`). Added a branch that unwraps the dict.

## 4.4 Results and confidence signals

| Unknown | Type | Predicted ID | Similarity | Margin (1→2) | Confidence signal |
|---|---|---|---|---|---|
| ID1 | ArcFace (512-d) | **22** | 0.664 | — | High — ArcFace identify() top-1 |
| ID2 | ArcFace (512-d) | **8** | 0.580 | — | High |
| ID3 (updated) | ArcFace (512-d) | **37** | 0.700 | — | High |
| ID4 | VGG (1000-d logits) | **24** | 0.734 (centroid) / 0.270 (centered+max-sim) | 0.034 / 0.017 | Moderate — two methods agree |
| ID5 (updated) | VGG (1000-d logits) | **12** | 0.859 (centroid) / 0.408 (centered+max-sim) | 0.019 / 0.036 | Moderate — two methods agree; #3 drops 25% |

**Final submission. 5 / 5** predictions produced. The 3 ArcFace matches are high-confidence; the 2 VGG matches are supported by *agreement between two independent scoring schemes* (centroid

on raw logits, and max-similarity on mean-centered logits) — if the features were noise, the two methods would land on different classmates.

**Honest caveat on VGG confidence.** A k-NN vote over the top-10 nearest gallery images gives only 2/10 agreement for both VGG unknowns — consistent with VGG19's 1000-d ImageNet logits not being face-discriminative at the resolution we need. With 1–3 photos per classmate and 38 classes to discriminate, even a 2/10 vote only barely beats chance (≈1/38 = 2.6/10). The centroid/max-sim agreement is the strongest confidence we can extract from this feature set; better VGG signal would require more photos per classmate, alignment (not just bbox crop), and/or fc2 features instead of pre-softmax logits (all ruled out by the input format the TA provided).

### 4.5 Where to look in the code

| Artifact | Location |
|---|---|
| ArcFace matcher | `scripts/identify_unknowns.py` |
| VGG logits matcher | `scripts/match_vgg_logits.py` (flags: `--classmates-only`, `--center`, `--max-sim`, `--knn`) |
| Challenge notebook | `notebooks/Challenge_Day.ipynb` |
| Walker + parser patches | `src/feature_extraction.py::walk_image_tasks`, `src/metadata.py::parse_base_identity` |

## 5. Appendix: Full Classifier Comparison

Reproduced verbatim from Stage 3 notebook cell 10 and `scripts/evaluate.py` output. All accuracies on the 20% held-out test set.

| Task | Classifier | Overall | Clean | Expression | Lighting | Occlusion | Side |
|---|---|---|---|---|---|---|---|
| Condition | ArcFace — LR | 0.545 | 0.656 | 0.462 | 0.375 | 0.833 | 0.500 |
| Condition | VGG19 — LR | **0.612** | 0.538 | 0.500 | 0.559 | 0.917 | 0.857 |
| Identity | ArcFace — LR | **1.000** | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Identity | VGG19 — LR (raw) | 0.810 | 0.718 | 0.955 | 0.882 | 0.750 | 0.714 |
| Identity | VGG19 — L2 + PCA + LR | 0.785 | 0.744 | 0.864 | 0.824 | 0.667 | 0.786 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Identity | VGG19 — MLP head | 0.818 | 0.744 | 0.818 | 0.912 | 0.750 | 0.857 |
| Identity | VGG19 — SupCon + NN centroid | produced by `evaluate.py` step [7]; saved to `artifacts/vgg19_contrastive_head.pth` | | | | | |
| Identity | ArcFace `identify()` cosine | **1.000** | 97 / 97 probe images correct | | | | |

## Deliverables checklist

| Rubric item | Where | Status |
|---|---|---|
| LR condition classifier on ArcFace/VGG | `src/train.py::train_condition_classifier` | done (0.545 / 0.612) |
| Condition performance across conditions + failure analysis | §2.1, §2.3 + `evaluate.py` step [9] | done |
| Train vs test gap analysis | §2.1 last paragraph, §2.3 last paragraph | done |
| VGG finetune: LR-on-top | `train_vgg_pca_pipeline` (0.785) | done |
| VGG finetune: layers (MLP head + contrastive) | `train_vgg_mlp_head` (0.818), `train_vgg_contrastive_head` | done |
| Failure cases for VGG identity | §2.3 per-condition table (Side/Occlusion gap) | done |
| System documentation | §3 + `README.md` | done |
| TA reproduction instructions | §3.3 ( `build_gallery.py` + `evaluate.py` ) | done |
| 5 unknown identity vectors matched | §4.4 | 5/5 produced |
| ≥3 identities correct (full credit) | §4.4 — 3 ArcFace high-confidence + 2 VGG moderate | targeted |